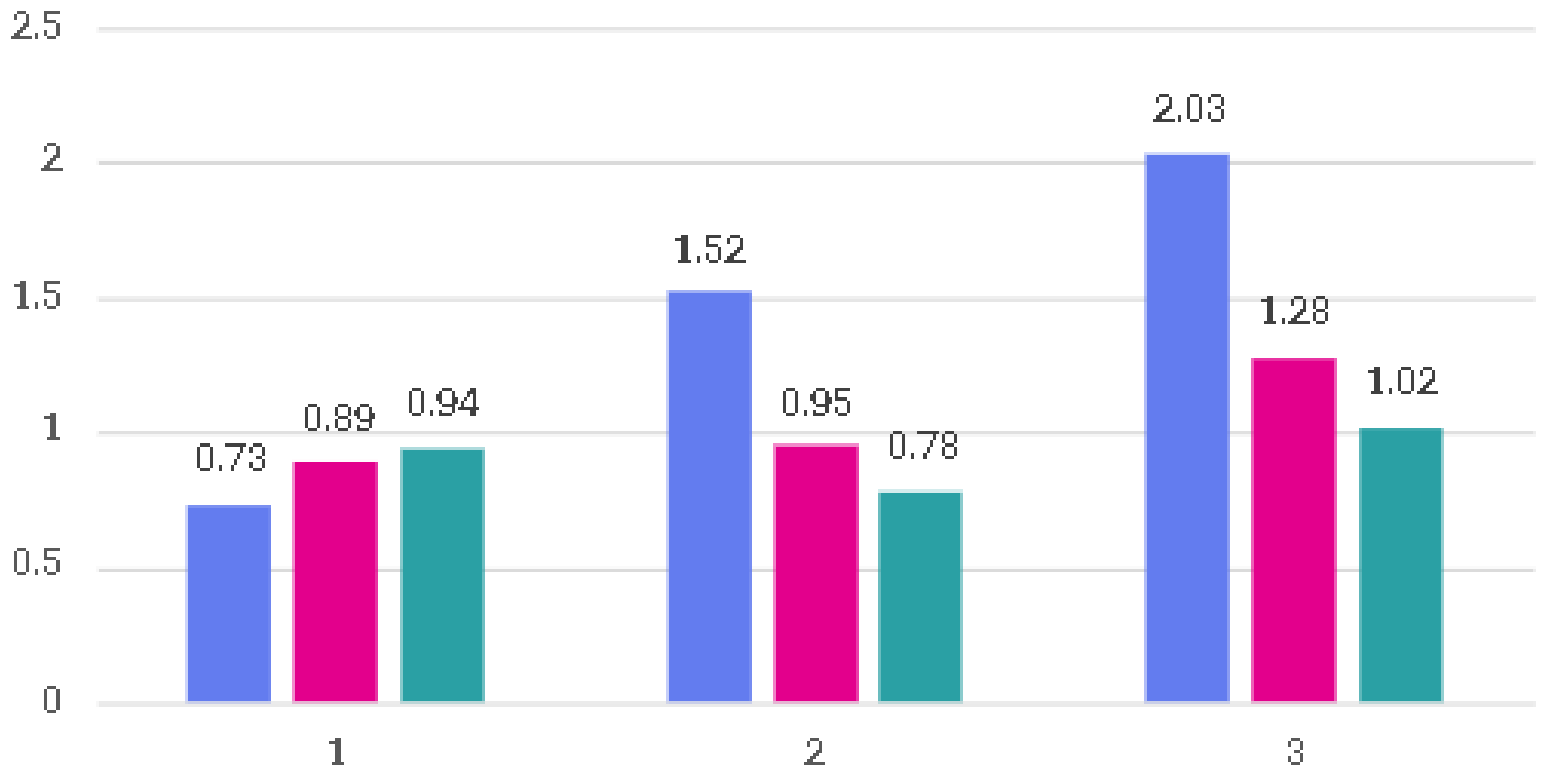


Top 10 Food Products by Emissions



Chart Title

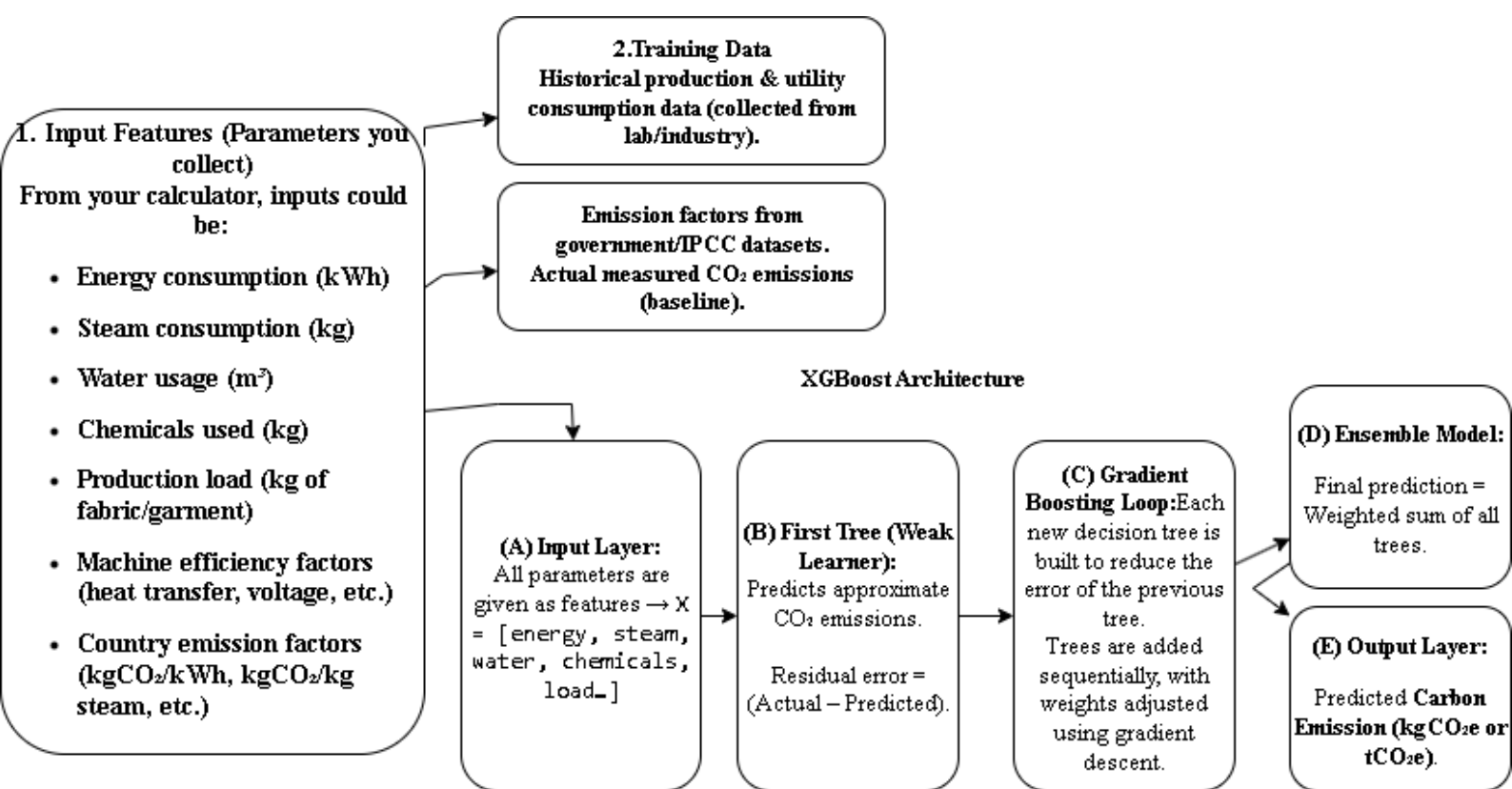
Linear Regression Random Forest XGBoost (Final Model)

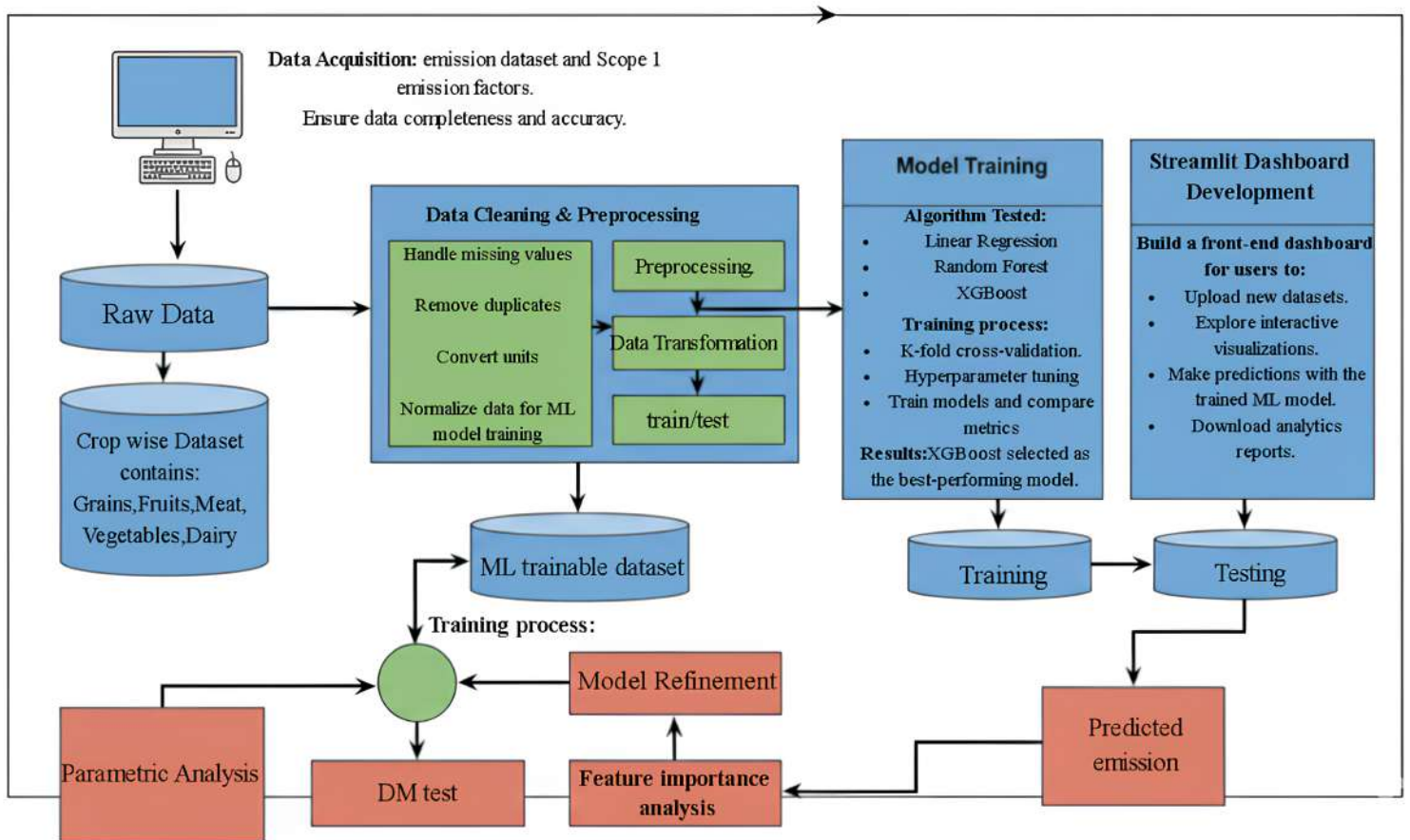


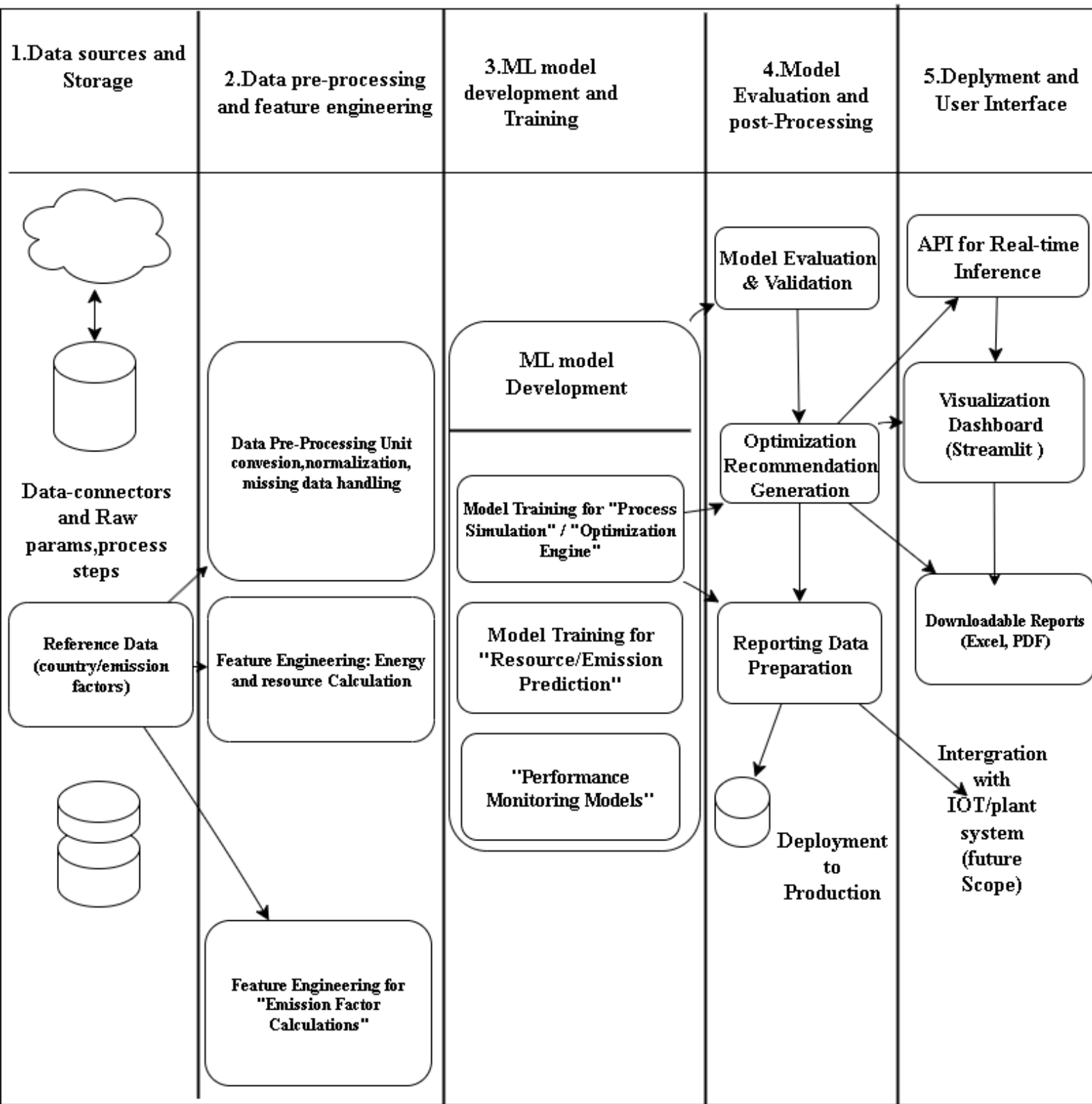
| Algorithm             | $R^2$ Score | MAE         | RMSE        |
|-----------------------|-------------|-------------|-------------|
| Linear Regression     | 0.73        | 1.52        | 2.03        |
| Random Forest         | 0.89        | 0.95        | 1.28        |
| XGBoost (Final Model) | <b>0.94</b> | <b>0.78</b> | <b>1.02</b> |

| Feature Name       | Description                                        | Unit                      |
|--------------------|----------------------------------------------------|---------------------------|
| Food Product       | Name of the food item                              | -                         |
| Land Use Change    | Emissions from land-use changes like deforestation | kg CO <sub>2</sub> e/kg   |
| Feed               | Emissions from animal feed production              | kg CO <sub>2</sub> e/kg   |
| Farm               | On-farm emissions (fertilizers, irrigation, etc.)  | kg CO <sub>2</sub> e/kg   |
| Processing         | Emissions during food processing                   | kg CO <sub>2</sub> e/kg   |
| Transport          | Transportation emissions                           | kg CO <sub>2</sub> e/kg   |
| Packaging          | Packaging material-related emissions               | kg CO <sub>2</sub> e/kg   |
| Retail             | Emissions at retail stage (storage, refrigeration) | kg CO <sub>2</sub> e/kg   |
| Transport EF       | Emission factor for transport                      | kg CO <sub>2</sub> e/unit |
| Sum Components     | Sum of all lifecycle stage emissions               | kg CO <sub>2</sub> e/kg   |
| Transport Fraction | Ratio of transport emissions to total              | - (percentage)            |

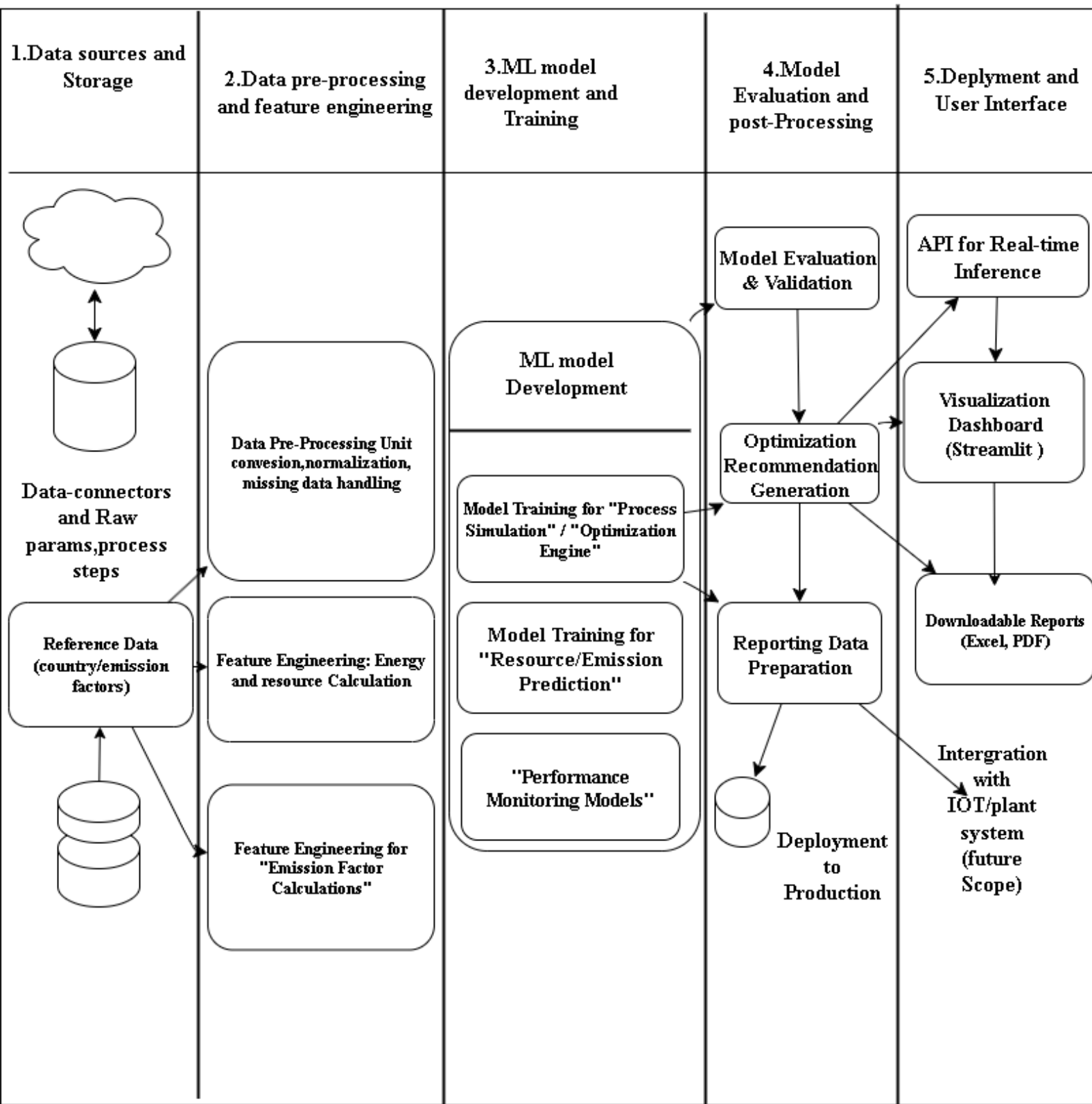
| Metric                                              | Formula                                                                      | Purpose                                                      |
|-----------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>R<sup>2</sup> (Coefficient of Determination)</b> | $R^2 = 1 - \frac{\sum (Y_{true} - Y_{pred})^2}{\sum (Y_{true} - \bar{Y})^2}$ | Measures how well predictions fit the actual data.           |
| <b>MAE (Mean Absolute Error)</b>                    | $MAE = \frac{1}{n} \sum  Y_{true} - Y_{pred} $                               | Represents the <b>average magnitude of prediction errors</b> |
| <b>RMSE (Root Mean Squared Error)</b>               | $RMSE = \sqrt{\frac{1}{n} \sum (Y_{true} - Y_{pred})^2}$                     | Penalizes larger errors more significantly.                  |

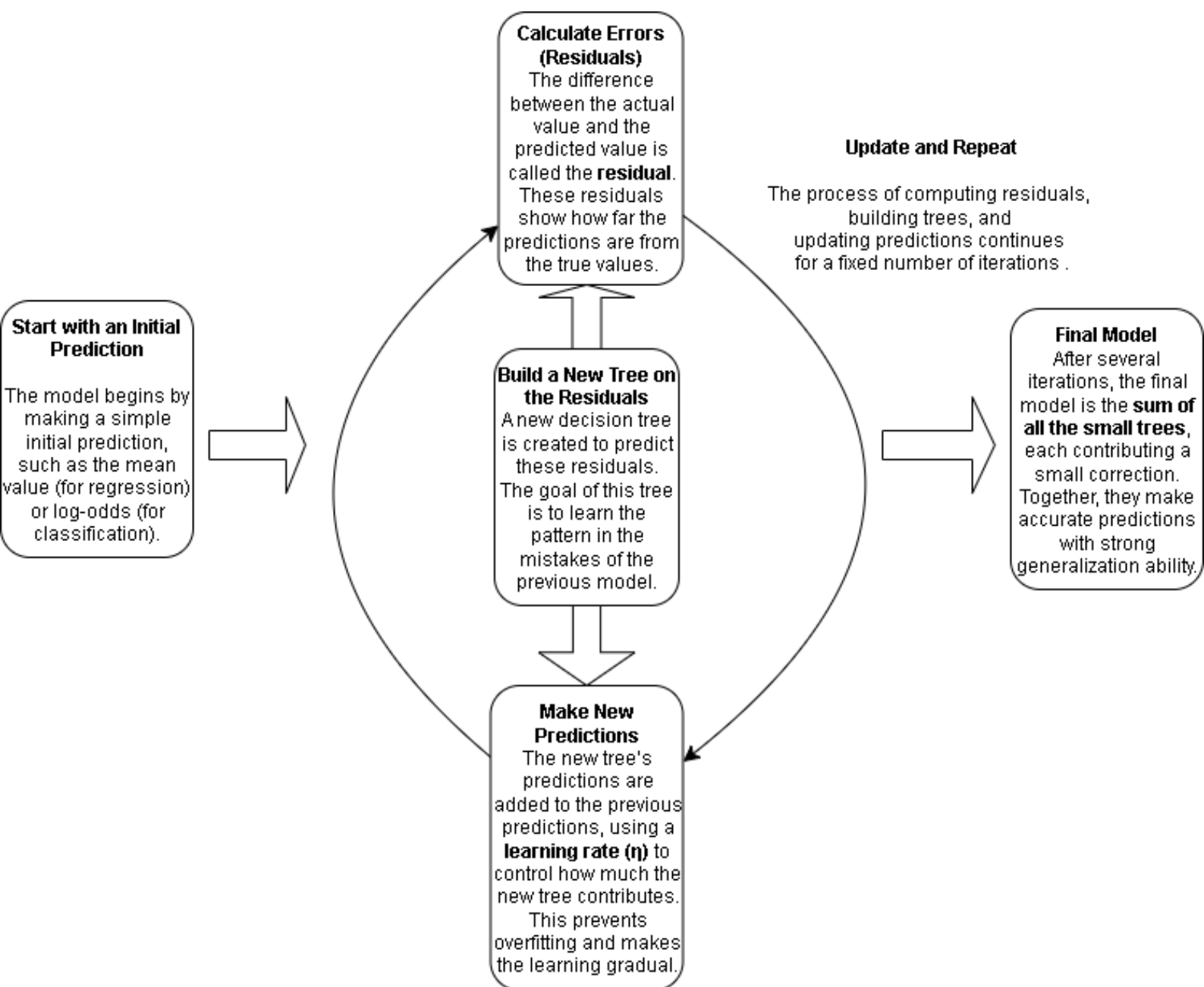


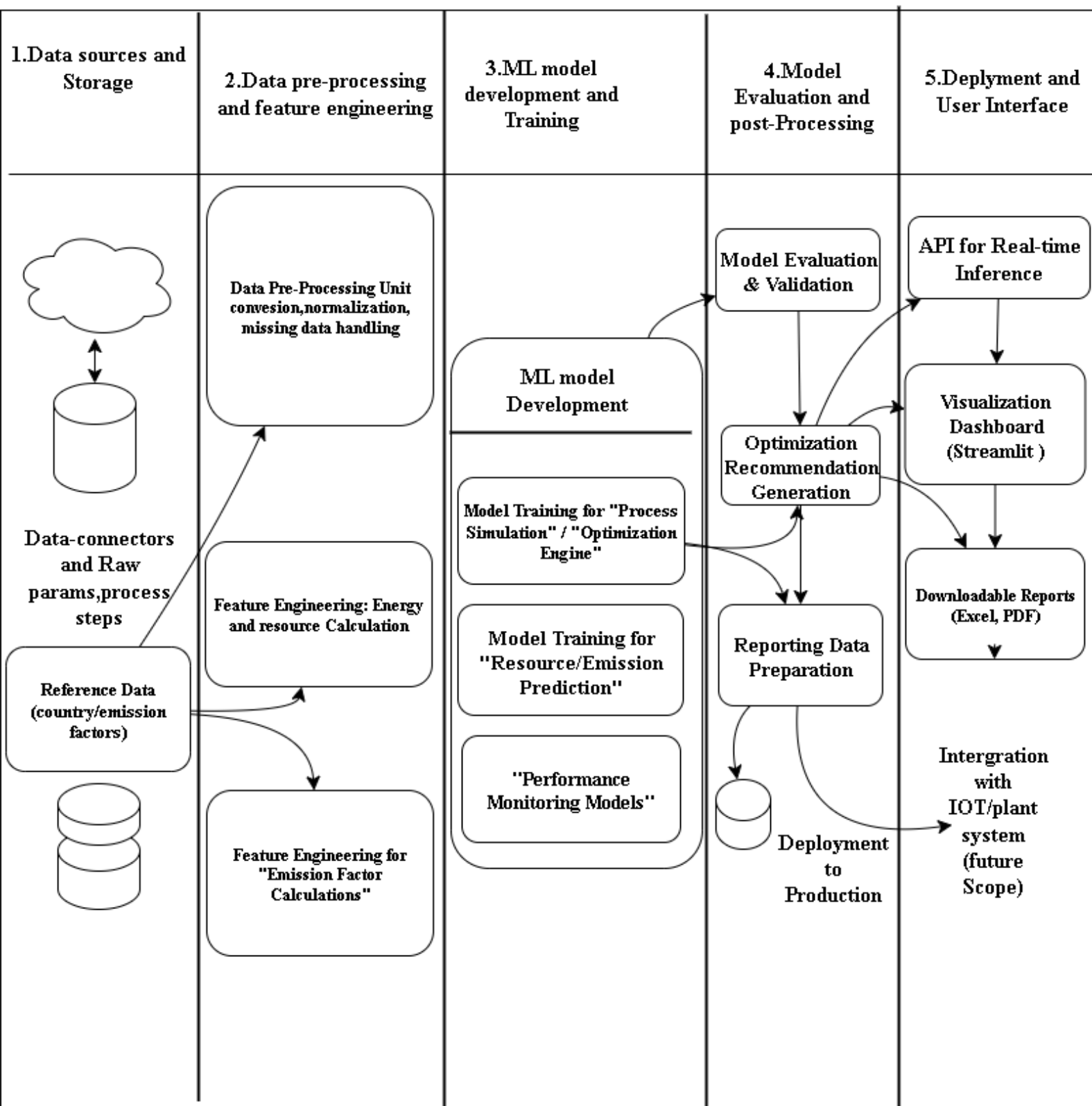


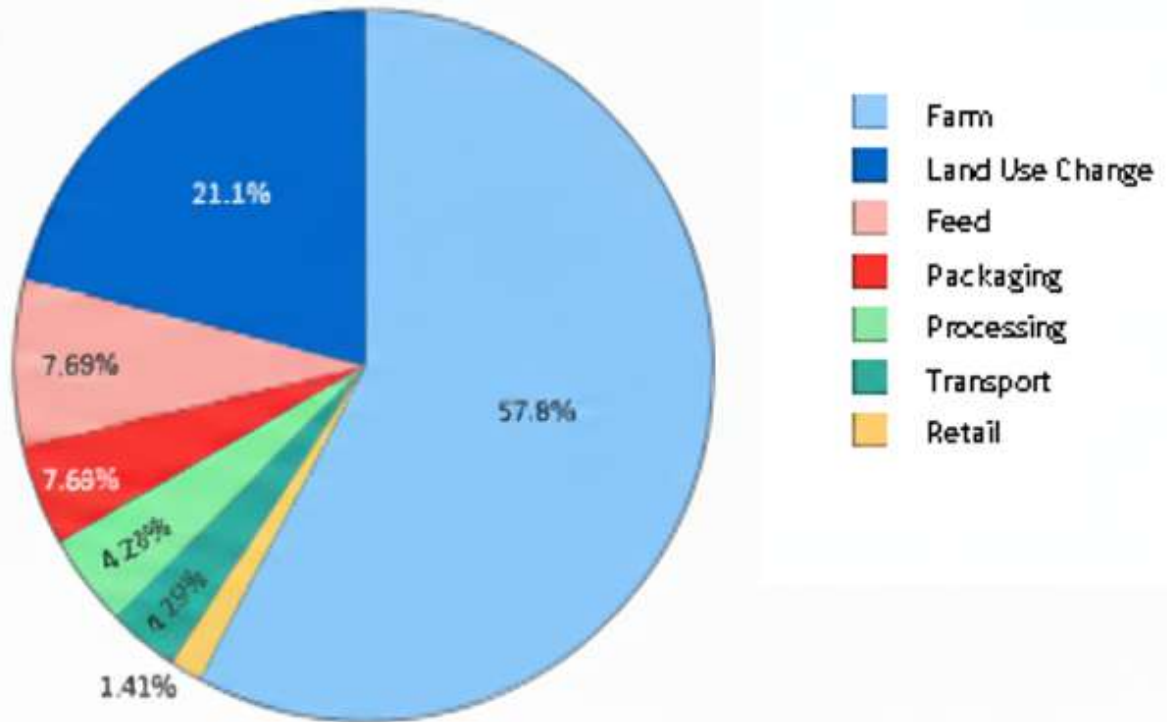












Lifecycle Stage Contribution per Top Product

